



Data-driven toxicity prediction in drug discovery: Current status and future directions

Ningning Wang^{1,2,3}, Xinliang Li^{1,2,3},
Jing Xiao⁴, Shao Liu^{1,2,3,*}, Dongsheng Cao^{1,2,5,*}

¹ Department of Pharmacy, Xiangya Hospital, Central South University, Changsha 410008 Hunan, PR China

² National Clinical Research Center for Geriatric Disorders, Xiangya Hospital, Central South University, Changsha 410008 Hunan, PR China

³ The Hunan Institute of Pharmacy Practice and Clinical Research, Changsha 410008 Hunan, PR China

⁴ Hunan Institute for Drug Control, Changsha 410001 Hunan, PR China

⁵ Xiangya School of Pharmaceutical Sciences, Central South University, Changsha 410013, Hunan, PR China

Early toxicity assessment plays a vital role in the drug discovery process on account of its significant influence on the attrition rate of candidates. Recently, constant upgrading of information technology has greatly promoted the continuous development of toxicity prediction. To give an overview of the current state of data-driven toxicity prediction, we reviewed relevant studies and summarized them in three main respects: the features and difficulties of toxicity prediction, the evolution of modeling approaches, and the available tools for toxicity prediction. For each part, we expound the research status, existing challenges, and feasible solutions. Finally, several new directions and suggestions for toxicity prediction are also put forward.

Keywords: toxicity prediction; machine learning; deep learning; predictive tools

Introduction

Drug research and development (R&D) is a 'knowledge-concentrated' process characterized by high risk, high investment, and long periods. Although much progress has been made in pharmaceutical technology, it is estimated that the attrition rate of drug candidates remains high, at 96%, with 30% of the attrition being due to a lack of safety.^{(p1),(p2),(p3),(p4)} There are countless



Ningning Wang is a pharmacist-in-charge of Xiangya Hospital, Central South University, China. She received her bachelor's degree and master's degree from Central South University, and currently focuses on artificial intelligence-assisted drug safety evaluation and clinical pharmacy.



Shao Liu is a doctor of pharmacy, postdoctor of clinical medicine. Currently, he is a chief pharmacist, professor, and doctoral supervisor of the Department of Pharmacy of Xiangya Hospital, Central South University, China. His research interests mainly include pharmaceutical administration, safe and rational use of drugs, and new drug development.



Dongsheng Cao is currently a professor at the Xiangya School of Pharmaceutical Sciences, Central South University, China. He received his PhD in 2013 from Central South University, China. His research interests include (1) artificial intelligent systems for drug discovery and disease diagnosis; (2) the development of software, Web service, and database in systems biology and drug discovery; (3) design and discovery of small molecular inhibitors of important protein targets. Further information about Dong-Sheng Cao can be found at the website of his group: <http://www.scbdd.com>.

potential entities in clinical trials, and even after marketing some have had to be withdrawn due to unacceptable toxicity. Environmental science is another area affected by the toxicity of chemical compounds, as any substance must meet the toxicity test standards of regulatory agencies before it is put into use to ensure that it does not cause harm to the environment and organisms. Therefore extensive safety evaluation of new chemical entities

* Corresponding authors. Liu, S. (liushao999@csu.edu.cn), Cao, D. (oriental-cds@163.com).

should be conducted as early as possible in the drug development and environmental safety evaluation process.^(p5)

Early safety assessment mainly relies on traditional *in vivo* and *in vitro* methods, which are limited in terms of application due to inherent shortcomings such as expense, ethical restrictions, and their time- and labor-consuming nature.^{(p6),(p7),(p8)} Subsequently, with the popularity of computational technology and interdisciplinary methods, *in silico* methods such as structural alerts, read-across, and analysis of quantitative structure–activity relationships (QSARs) have shown unique advantages for toxicity assessment in the early stages of drug design, including their green, fast, cheap, and accurate nature; most importantly, they can be applied before a compound is synthesized.^{(p9),(p10)} Regarding QSAR analysis, since this scientific concept was formally proposed in 1962,^{(p11),(p12)} various modeling methods have been developed to process the increasing amount of toxicity data. Among them, machine learning (ML) has become the fastest-growing and most widely used method, serving as an important component of data-driven toxicity prediction research.^{(p5),(p8),(p13),(p14),(p15)} Generally, data-driven toxicity prediction studies using ML aim to build classification or regression models to describe the complex relationships between the chemical structure of molecules and their toxicity endpoints based on knowledge obtained from experimental data.^{(p16),(p17),(p18)} Continuous toxicity prediction studies and several reviews in the past few years have confirmed the utility of ML methods in the field of toxicity prediction.^{(p5),(p8),(p14),(p15)} Furthermore, recently, owing to hardware improvements and the availability of extremely large datasets, deep learning (DL), a subfield of ML, has surpassed traditional ML methods in the toxicity assessment field.^{(p19),(p20),(p21)}

In recent years, constant upgrading of information technology has promoted the continuous development of toxicity prediction.^{(p8),(p22),(p23)} To get a full picture of data-driven toxicity prediction in drug discovery and environmental science, we review the available ML-based studies in this paper and summarize them in the following three respects [Figure 1](#). (1) The features and difficulties of data-driven toxicity prediction, including small datasets, complex mechanisms, and low-quality data. (2) The evolution of modeling methods, from traditional ML to DL and the post-DL era, mainly focusing on the advantages and disadvantages of various learning methods, deep relationships, and the differences between methods. (3) Popular software and tools for toxicity prediction, mainly discussing the endpoints covered, modeling methods, prediction accuracy, and explanatory ability. For (1)–(3), the existing issues and feasible suggestions to overcome them are discussed.

The features and difficulties of data-driven toxicity prediction

In the past few decades, computer-aided drug design (CADD) has undergone rapid development, and toxicity evaluation, as an important part of this, has also attracted extensive attention. With the continuous improvement of computing power and the proposal of new theories, more and more CADD researchers have attempted to derive effective methods to construct

regression or classification models for a variety of toxicity endpoints and have achieved some satisfactory results. However, for some toxicity endpoints, prediction performance is consistently at a lower level even though various ML methods have been used. To identify the underlying reasons and comprehensively understand the features and difficulties of data-driven toxicity prediction, we collated 45 toxicity endpoints that had previously been predicted and list their information in [Figure 2](#). Based on this figure, we summarize several features for toxicity prediction tasks as follows.

Relatively small datasets

To explore the underlying features of toxicity datasets, we collected 50 datasets covering 45 different endpoints and drew a pie diagram to represent them ([Figure 2a](#)). From this figure, we can see that only 34% of datasets (17/50) contain more than 5000 compounds, and only one dataset (hERG) has more than 10,000 compounds. Specifically, most of the datasets (12/17) containing sufficient compounds (>5000) are pathway toxicity datasets [human ether-a-go-go related gene (Tox21) datasets],^{(p24),(p25),(p26),(p27)} and the remaining datasets generally correspond to a class of toxicity endpoints represented by hERG blockers,^{(p28),(p29),(p30),(p31)} Ames mutagenicity toxicity,^{(p32),(p33),(p34),(p35),(p36),(p37)} cytotoxicity,^{(p38),(p39),(p40),(p41)} and oral rat acute toxicity.^{(p42),(p43),(p44),(p45)} Owing to their unique toxicological significance, definite endpoints, relatively simple mechanism, and sufficient experimental data, the datasets were selected for studying, at an early stage, the application of ML to toxicity prediction, and subsequent intensive research made them well-known in this field. However, most toxicity datasets are relatively small (<5000, 33/50), with a large proportion having fewer than 500 compounds (11 of 50, 22%). In recent years, with the release of more and more toxicity databases, the data on some toxicity endpoints have been slightly supplemented (e.g. developmental toxicity,^{(p46),(p47),(p48)} hepatotoxicity,^{(p49),(p50),(p51)} and genotoxicity^{(p52),(p53),(p54)}), but there are still some complex toxicity endpoints lacking data, such as nephrotoxicity,^{(p55),(p56)} rhabdomyolysis,^(p57) hematotoxicity,^{(p58),(p59)} and so on.^{(p60),(p61),(p62),(p63),(p64),(p65)} From the perspective of mathematical modeling, a smaller dataset carries less information, resulting in poor predictive ability of the output model; this cannot be significantly improved even by using advanced modeling methods, and several recent studies using different modeling methods have shown that DL does not perform as well as expected in predicting some specific toxicity endpoints.^{(p50),(p58),(p65)} In summary, a relatively small dataset greatly limits the prediction of toxicity endpoints, which is a major obstacle to the development of current and future data-driven toxicity assessments.

Complex mechanism

From the point of view of modern medicine, most toxicity endpoints have complex mechanisms of action, involving physiology, pathology, and pharmacology. The complexity of the mechanisms is reflected in two aspects: unknown mechanisms and multiple mechanisms. Toxicity endpoints with unknown mechanisms are usually too heterogeneous and complex, lacking

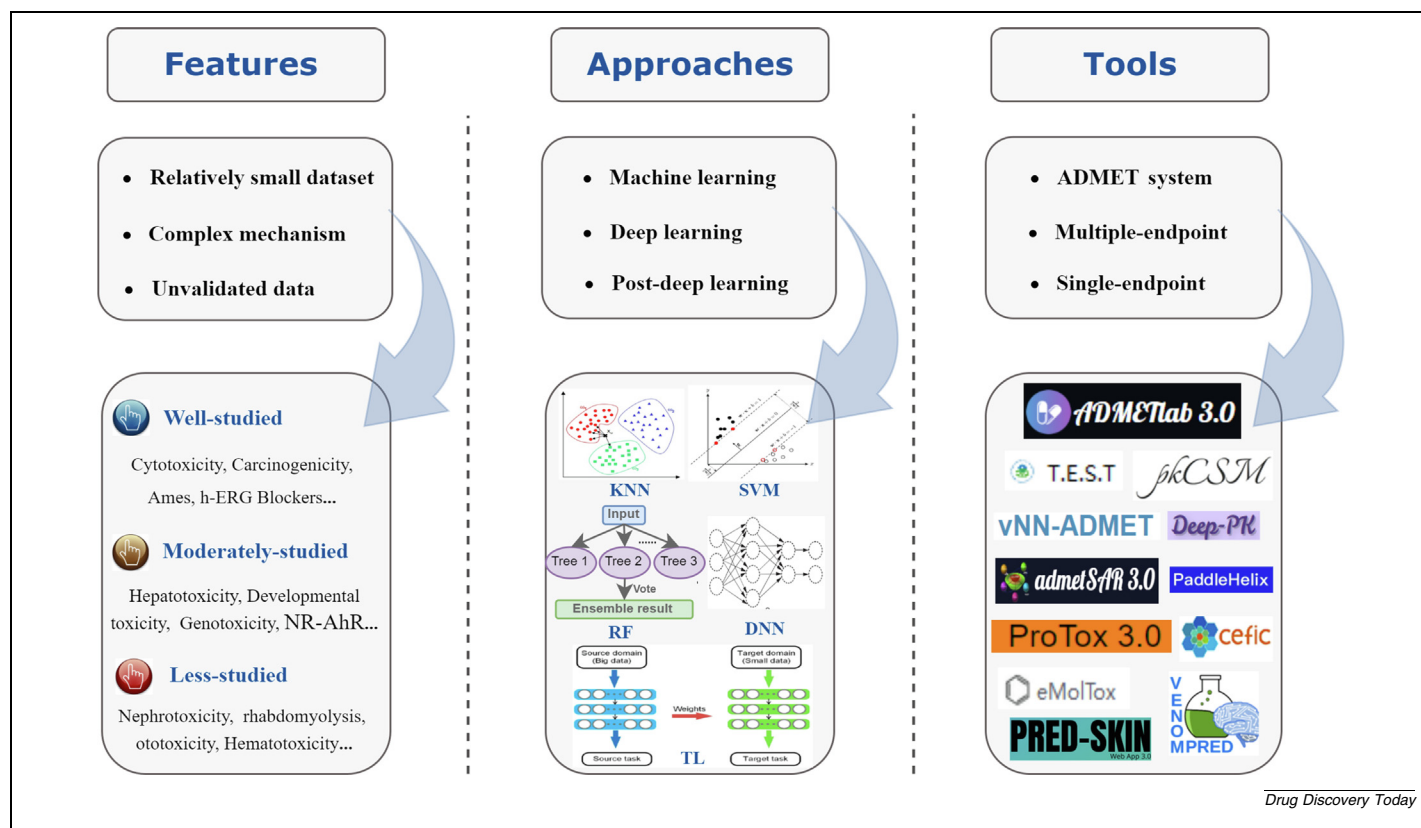


FIGURE 1

Overview of the data-driven toxicity prediction. This schematic consists of three parts. The first part pertains to the features and difficulties of data-driven toxicity prediction, including relatively small datasets, complex toxicity mechanisms, and unvalidated data. The second part pertains to toxicity prediction methods, including machine learning, deep learning and post-deep learning methods. Finally, currently available toxicity prediction tools are listed, including the ADMET system and prediction tools for specific toxicities.

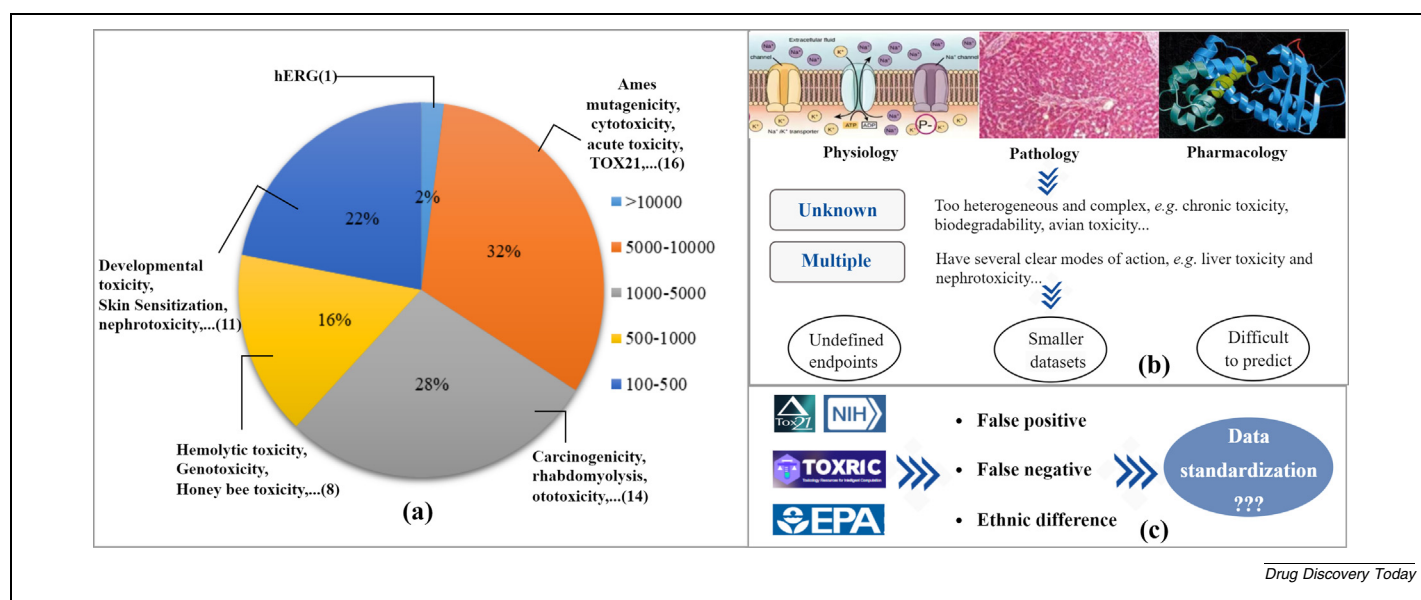


FIGURE 2

Features of toxicity prediction (a) Relatively small datasets. The pie diagram shows that most of the toxicity datasets are relatively small (<5000, 33/50), and some contain fewer than 500 compounds (11 of 50, 22%). (b) The complex mechanism of chemical toxicity. Most toxicity endpoints have complex mechanisms of action encompassing physiology, pathology, and pharmacology, which is reflected in two aspects: unknown mechanisms and multiple mechanisms; (c). Existing problems with toxicity data, including false-positive data, false-negative data, and the differences between animal tests and human data, have led to bias in toxicity predictions.

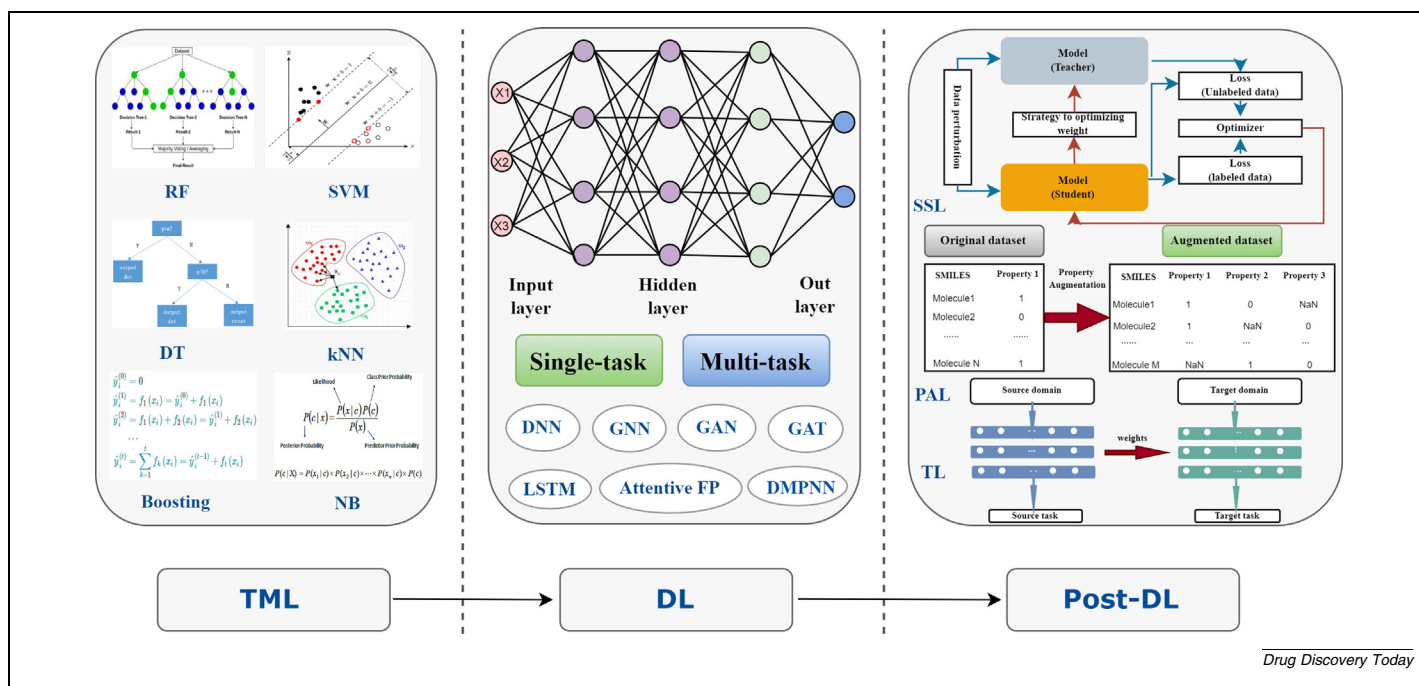


FIGURE 3

Computational methods commonly used in toxicity prediction. There are three main categories: traditional machine learning, including random forest, support vector machine, and decision tree; deep learning, including deep neural networks, convolutional neural networks, and recurrent neural networks; and the post-deep learning era, including semi-supervised learning, transfer learning, and property augmentation learning.

an easily understandable mechanism or to the ability to be well explained, such as in terms of chronic toxicity, biodegradability, avian toxicity, etc.^{(p25),(p60),(p61),(p66)} The other aspect is toxicity endpoints with multiple mechanisms, which have several well-defined modes of action, represented by liver toxicity and nephrotoxicity.^{(p50),(p51)} Taking hepatotoxicity as an example, its computational prediction is still extremely challenging due to the complex underlying mechanisms, such as direct hepatocytotoxicity, immune-related damage, mitochondrial damage, and the inflammatory response.^{(p49),(p68)} Therefore although a great deal of work has been done to try to accurately predict drug-induced liver injury (DILI), the results are not very satisfactory. Generally, the complex mechanism of toxicity endpoints will lead to undefined endpoints and smaller datasets, which will further result in increased prediction difficulty and decreased predictive ability. Therefore the data-driven prediction of some toxicities is a difficult task due to their inherently complex mechanisms. Based on the current research, we hold the opinion that collecting more well-validated data and developing more elaborate prediction models with distinct toxicity mechanisms or endpoints may be a feasible direction for the future prediction of toxicity endpoints with complex mechanisms.

Low-quality data

For data-driven toxicity prediction, the quality of the data is still a serious and fundamental issue. At present, most available toxicity data are collected through high-throughput *in vitro* assays or *in vivo* tests on animals, such as Tox21 and ToxCast.^(p69)

Although they provide an opportunity for further development of toxicity prediction based on ML, it is undeniable that there are still some problems including false-positive data, false-negative data, and the differences between animal and human data, which introduce bias into toxicity prediction. This is the main reason why there is a large difference between the results predicted by most QSAR models and the clinical reality. To improve the problem of low quality toxicity data, one obvious method is to collect more reliable clinical data, and another effective solution is to standardize the available toxicity data. Although there are already many databases containing clinical toxicity data, such as the SIDER, LiverTox, PharmGKB, and [Drugs.com](https://www.drugsc.com) databases, we need an open-access benchmark database with clear and accurate descriptions of toxicity endpoints and even the adverse outcome pathway (AOP) of toxicity. The quality of toxicity data from laboratories in different countries and regions is highly variable, and the low interlaboratory repeatability of the data will lead to a phenomenon whereby, as modeling methods improve, the prediction accuracy of a given model will reach saturation such that, even if more test data can be accumulated, further improvement in predictive power is difficult.^{(p70),(p71),(p72),(p73),(p74),(p75)} This is also a major problem in terms of current data collection, which fundamentally hinders the continuous optimization of toxicity prediction models. Therefore from the point of view of toxicity data, there are two main future directions: (1) collect and screen more high-quality clinical toxicity data to improve the practicability of the data; and (2) develop a standardized process for toxicity

data validation and establish a large benchmark database consisting only of well-validated test results to reduce interlaboratory bias.

Modeling approach for toxicity prediction

In the process of data-driven toxicity prediction, the choice of modeling approach is a central and crucial issue. Over the past few decades, the development of computer technology has facilitated the emergence of various modeling approaches, which are roughly classified into three types/stages in this review: traditional machine learning (TML), DL, and the post-deep learning era (PDL). Different modeling methods are derived from different practical requirements and have different characteristics and application fields [Figure 3](#).

TML

In the early stage of toxicity prediction, statistical analysis methods, such as multiple linear regression (MLR) and partial least squares (PLS), were first employed to explore the possible relationships between chemical structures and toxicities, but the complexities of biological systems and the increasing volume of experimental data make these methods unsatisfactory in many cases. At this time, TML attracted the attention of toxicologists as a class of advanced algorithms that can model more complex tasks. Popular TML methods in toxicity prediction studies mainly include support vector machine (SVM), random forest (RF), decision tree (DT), k-nearest neighbor (kNN), naïve bayes (NB), etc. Their basic concept, statistical characteristics, advantages, and disadvantages have been fully discussed in several previous reviews. [\(p5\),\(p8\),\(p14\),\(p15\)](#)

In recent years, with the increasing maturity of TML methods and the continuous increase in toxicity data, researchers have compared these popular TML methods to find the one most suitable for each particular toxicity. [\(p46\),\(p52\),\(p55\),\(p56\),\(p57\),\(p61\),\(p62\),\(p63\),\(p66\),\(p67\),\(p76\)](#) We briefly summarize their details in [Table 1](#), including 12 toxicity endpoints, the TML methods applied, the optimal method, the predictive ability of the final model, the application area, and the original references. Among these methods, SVM performed best for most toxicity endpoints, followed by RF. This seems intuitive. SVM, also known as support vector classifier (SVC) or support vector regression (SVR) in the context of particular tasks, is well known for its high predictive performance and low risk of overfitting. In the context of toxicity prediction, the SVM algorithm uses kernel functions, such as linear, polynomial, and sigmoid functions, as well as the radial basis function (RBF), to project molecules encoded by descriptor vectors into a space that maximizes the margin (or separation/boundary) between different classes. [\(p77\)](#) Considering the difficulties of toxicity prediction associated with small datasets, complex mechanisms, and mixed information, SVM, as a supervised TML algorithm to handle datasets with high-dimensional variables, will be the optimal solution in many cases. As for RF, an ensemble modeling method based on building multiple DTs as the base learner is a nonlinear multiclass ML method. [\(p15\)](#) Similar to SVM, one of the main advantages of RF is that it can handle a large number of features without overfitting and is insensitive to data with missing values and noise. Other TML methods, including naïve Bayes (NB), kNN, the artificial neural network (ANN), extreme gradient boosting (XGBoost), etc., have also been applied to model toxicity data in these studies, but their performance is not as satisfactory. To sum up, although

TABLE 1
Comparison of different TML methods for toxicity prediction

Toxicity endpoint	Studied TML methods	Optimal TML method	Internal validation	External validation	Application area	Refs
Genotoxicity	SVM, NB, kNN, DT, RF, ANN	SVM	ACC = 0.889	ACC = 0.908	Drug discovery	(p52)
Urinary tract toxicity	RVM, SVM, RF, DT, XGBoost, AdaBoost, SVMBoost, RVMBoost	SVMBoost	AUC = 1.000 q _{LOO} ² = 0.904	AUC = 0.893; q _{ext} ² = 0.845	Drug discovery	(p56)
Rhabdomyolysis	kNN, ANN, SVM, fast stagewise multivariate linear regression, MLR, PLS, RF, deep neural network	RF	ACC = 0.793	ACC = 0.753	Drug discovery	(p57)
Avian toxicity	SVM, DT, kNN, RF, NB	SVM	ACC = 0.866	ACC = 0.851	Environmental safety	(p61)
Hemolytic toxicity	kNN, SVM, RF, GBM	SVM	Q ² = 0.653	–	Drug discovery	(p62)
Mitochondrial toxicity	kNN, LR, RF, SVM, XGBoost	RF	AUC = 0.899	AUC = 0.921	Drug discovery	(p63)
Honey bee toxicity	ANN, DT, kNN, NB, RF, SVM	SVM	ACC = 0.890	ACC = 0.904	Environmental safety	(p64)
Biodegradability	SVM, DT, kNN, NB	SVM	AUC = 0.891	AUC = 0.844	Environmental safety	(p66)
Nephrotoxicity	ANN, SVM	SVM	ACC = 0.969	ACC = 0.933	Drug discovery	(p67)
Aquatic toxicity	kNN, SVM, ANN, RF, NB, DT	SVM	ACC = 0.760	ACC = 0.850	Environmental safety	(p76)
Neurotoxicity	BGR, ETR, GPR, kNN, MLPR, Nu-SVR, RF, SVR	ETR	q _{test} ² = 0.828	q _{test} ² = 0.784	Drug discovery	(p55)
Developmental toxicity	NB, SVM, RP, kNN, DT, RF, AdaBoost	NB	ACC = 0.935; Q ² = 0.728	ACC = 0.891; R ² = 0.697	Drug discovery	(p46)

Abbreviations: ACC, accuracy; ANN, artificial neural network; XGBoost, extreme gradient boosting; GBM, gradient boosting machine; LR, logistic regression; BGR, bagging regressor; ETR, extra trees regressor; GPR, Gaussian process regression; RP, recursive partitioning; RVM, relevance vector machine.

TABLE 2

Detailed information on representative toxicity prediction studies based on DL

Category	Toxicity endpoint	DL framework	Internal validity	External validity	Application area	Refs
Single task	Tox21 (7 NR, 5 SR)	GCNN	AUC = 0.687–0.830	AUC = 0.757	Drug discovery	(p27)
	Liver toxicity	DCNN	SE = 0.790; SP = 0.520	AUC = 0.670	Drug discovery	(p49)
	Hepatocellular hypertrophy	DNN	AUC = 0.830	AUC = 0.720	Drug discovery	(p50)
	Hematotoxicity	GCNN, DMPNN, attentive FP	AUC = 0.768	AUC = 0.759	Drug discovery	(p59)
	Ototoxicity	TEXTCNN, transformer CNN	AUC = 0.930/0.940	AUC = 0.900/0.870	Drug discovery	(p66)
	DILI	CNN	–	AUC = 0.960	Drug discovery	(p78)
	Seven types of human organ toxicity	DMPNN	AUC = 0.700–0.800	AUC = 0.640–0.920	Drug discovery	(p79)
	hERG	DNN, GCN	ACC = 0.812	ACC = 0.773	Drug discovery	(p80)
	hERG	DNN, LSTM	BACC = 0.800–0.840	BACC = 0.720–0.760	Drug discovery	(p81)
	Hematotoxicity	TEXTCNN algorithm, CNN fingerprint, transformer CNN	AUC = 0.630–0.730	AUC = 0.620–0.680	Drug discovery	(p58)
	hERG	DMPNN	AUC = 0.956	AUC = 0.810–0.849	Drug discovery	(p82)
	hERG	FCNN, GCNN, CNN	ACC = 0.860, AUC = 0.930	ACC = 0.746–0.810	Drug discovery	(p83)
	hERG	DNN	AUC = 0.872–0.889	ACC = 0.814–0.992	Drug discovery	(p84)
	DILI	GNN	–	ACC = 0.773	Drug discovery	(p85)
	DILI	GAN	ACC = 0.814	ACC = 0.787/0.765	Drug discovery	(p86)
	Mutagenicity	GCNN	ACC = 0.910, AUC = 0.950	ACC = 0.850, AUC = 0.880	Drug discovery	(p88)
	Mutagenicity	GCNN	AUC = 0.878	AUC = 0.838	Drug discovery	(p89)
	Carcinogenesis	GNN	–	MSE = 0.71; AUC = 0.730	Drug discovery	(p90)
	Acute toxicity	FFNN, GCNN, CNN	R ² = 0.652–0.866	R ² = 0.687–0.861	Drug discovery	(p92)
	Aquatic toxicity (three groups of organisms)	Transformer + DNN	R = 0.640–0.730	MAE = 1.97–6.04	Environmental safety	(p87)
Multitask	Chemical toxicity (27 endpoints)		AUC = 0.770–0.934	AUC = 0.886	Drug discovery	(p93)
	Skin corrosion/irritation	Attentive FP	AUC = 0.963	AUC = 0.891	Drug discovery	(p94)
	Tox21 (7 NR, 5 SR)	DNN	AUC = 0.700–0.930 AUC (NR.AR) = 0.350	–	Drug discovery	(p25)
	33 toxicity endpoints	Multitask graph attention framework	–	AUC = 0.618–0.981	Drug discovery	(p26)
	Ames mutagenicity	Multitask learning DNN	ACC = 0.730–0.900	ACC = 0.990	Drug discovery	(p32)
	Acute toxicity	Multitask DNN, Single-task DNN, RF, DL consensus architecture, GCNN	R ² = 0.652–0.866	R ² = 0.687–0.861	Drug discovery	(p92)
	Tox21 (7 NR, 5 SR)	Multitask DNN, multitask GCNN	AUC = 0.903, MCC = 0.580, Precision = 0.814	AUC = 0.897/0.912	Drug discovery	(p96)
	Tox21 (7 NR, 5 SR); clinical toxicity; RTECS <i>in vivo</i> toxicity	Multitask DNN	AUC = 0.991–0.994	AUC = 0.820–0.829	Drug discovery	(p97)
	hERG	Multitask DNN	AUC = 0.792–0.944	AUC = 0.883–0.967	Drug discovery	(p98)

Abbreviations: BACC, balanced accuracy; DCNN, deep convolutional neural network; FCNN, fully convolutional neural network; FFNN, feedforward neural network; MAE, mean absolute error; MLPR, multilayer perceptron regression; MSE, mean squared error; RTECS, Registry of Toxic Effects of Chemical Substances; SE, standard error; NR, nuclear receptor; SR, stress response.

various methods have been applied for toxicity prediction, SVM and RF have already become the two most important TML methods in data-driven toxicity prediction research in the current context based on their unique advantages, and this finding can serve an effective reference for subsequent related studies.

DL

With the proliferation of toxicity data and the continuous optimization of computing power, DL, as an important subset of ML, has become one of the most promising techniques in the field of toxicity prediction owing to its unique advantages that allow complex features to be constructed automatically.^{(p72),(p73)}

DL is also known as deep neural network (DNN) and is closely related to the ANN, a TML algorithm inspired by biological neural networks in the human brain. Unlike the ANN, which has only three layers (an input layer, a hidden layer, and an output layer), DL algorithms have more than one hidden layer and are therefore more suited for analyzing large amounts of toxicological data with complex mechanisms.^(p22) Through the layer-wise extraction and abstraction of features, DL algorithms can characterize data and learn statistical laws from massive training samples and can thus make more accurate predictions based on unknown data. Classical DL algorithms mainly include convolutional neural networks (CNNs), recurrent neural networks (RNNs), generative adversarial networks, and reinforcement learning (RL), which have made great progress in the fields of face/voice/pattern recognition, natural language processing, bioinformatics, and drug discovery.^(p78)

In recent years, many data-driven toxicity prediction studies have confirmed the notion that, when the training set is large enough, DNNs can perform better than ANNs and all the above-mentioned TML methods.^{(p25),(p26),(p49)} We classified studies using DL into two categories, single-task and multitask predictions, and list brief information thereon in Table 2. From this table, we can see that different DL algorithms have been proposed to construct toxicity prediction models, and they have yielded some satisfactory results. Popular DL frameworks in the field of toxicity prediction mainly include the DNN, graph attention network (GAN), graph neural network (GNN), graph convolutional network (GCN), long short-term memory (LSTM) network, directed message passing neural network (DMPNN), and attentive fingerprint (FP). A variety of toxicity endpoint prediction studies have benefited from this raft of DL approaches, the most representative of which are several well-studied toxicity endpoints with relatively abundant data such as Tox21 pathway toxicity,^(p79) hERG blocker toxicity,^{(p80),(p81)} hematotoxicity,^{(p58),(p59)} and so on. For single-task studies, DL methods have been applied to a specific toxicity endpoint, and most of the derived models have excellent predictive ability with evaluation metrics over 0.8. For example, as an important endpoint, the toxicity of hERG blockers has recently been predicted by most popular DL methods such as the DNN, GCN, LSTM, and DMPNN, obtaining satisfactory area under the receiver operating characteristic curve (AUC) values of up to 0.956 and 0.849 for internal and external validation, respectively.^{(p82),(p83),(p84)} Among them,

the GCN is a DL model for processing graph data. The traditional CNN is suitable for processing regular grid-type data, such as image data, but cannot deal well with irregular graph data, such as user-item relationships in social networks and recommendation systems. By extending the convolution operation to graph structure data, the GCN can effectively process nodes and edges in graph data. The most outstanding feature of the GCN is that the features it extracts are of sufficient quality even without training and using only randomly initialized parameters. LSTM is a type of RNN architecture designed to capture long-term dependencies in sequential data. Traditional RNNs can struggle to learn and retain information over long sequences due to the vanishing gradient problem, but LSTMs are specifically designed to address this issue by introducing a memory cell that can selectively learn, store, and forget information over time. For real-world application, LSTM networks are particularly effective in tasks that require the modeling of complex temporal patterns and long-range dependencies in data sequences. The MPNN is a class of supervised learning framework that can be applied to graphs. This framework abstracts some commonalities from the popular neural network models that support graph data, such as gate GNNs and graph attention networks. Unlike the traditional MPNN framework, where messages are passed from one node to its neighbors via the edges between them, the DMPNN is unique in that edges are directed, and edges, instead of atoms, act as the source and receiver of messages. These two features prevent information from being repeatedly passed back to its source, thus reducing noise. Recently, the DMPNN has been widely used for molecular property prediction because of its high efficiency and capacity for chemical representation.

In addition to the hERG blockers, in 2020, Kang *et al.* proposed a prediction model for DILI, named GLIT, using transcriptional response data and chemical structures.^(p85) GLIT learns the embedding vectors using the GAN, one of the most effective GNN architectures, which learns hidden representations of each node using the attention scores of a given node and its neighbors in a graph. In this architecture, a drug embedding vector is extracted from the last layer of a neural network with three layers, and a drug-induced gene expression embedding vector is extracted from a GNN using a biological knowledge graph called OmniPath. The two embedding vectors are concatenated with the dosage and duration of drug administration information, and fed to the prediction layers of GLIT to predict DILI. GLIT achieved an F1 value of 0.788, which is better than those baseline models. Also for DILI prediction, Huang *et al.* used a multi-view (MV)-GNN and appropriate augmentation strategy to obtain a DILI model with a cross-validation accuracy of 0.814 in 2021. The MV-GNN model can extract atom messages and bond messages simultaneously. It considers atom message passing and bond message passing as two parallel streams and allows the atom/bond messages to communicate during the passing phase. It takes advantage of the two perspectives and thus outperforms other baseline models trained on DILI datasets in extensive experiments.^{(p78),(p85),(p86)} Hematotoxicity, a serious but overlooked toxicity in drug discovery, has also attracted the attention of deep-learning researchers in recent years. In 2022,

Cao *et al.* constructed a prediction model based on attentive FP that yielded a balanced accuracy of 72.6% and an AUC value of 76.8% for the validation set.^(p59) Attentive FP is an attention mechanism that assigns different weights to different input features to pay more attention to target-relevant features and suppress target-irrelevant features. This mechanism is commonly used in neural networks to improve the performance and generalization of a model. Specifically, it can be applied in each layer of the neural network to enforce target-related information and suppress irrelevant or redundant information by dynamically redistributing weights. Attentive FP is proposed as a powerful attention mechanism that helps to improve the performance and generalization ability of neural networks, especially when dealing with toxicity data with complex relationships and many features. Recently, Kristiansson *et al.* presented an aquatic toxicity prediction model using transformers to capture toxicity-specific features directly from chemical structures and obtained high predictive performance. Their results have demonstrated that DL and transformers constitute an adaptive data-driven approach for estimating structure–toxicity relationships that improves the predictive performance of existing QSAR methods, which may have some new implications for data-driven toxicity prediction.^(p87) In addition to the toxicity properties mentioned above, these DL methods have also been successfully applied to some other toxicity endpoints, such as alternariol monomethyl ether (AME) toxicity,^{(p88),(p89)} carcinogenesis,^{(p90),(p91)} acute toxicity and so on.^{(p92),(p93),(p94)}

As well as single-task models, multitask prediction has also been vigorously developed in the context of DL.^(p95) Multitask learning is an ML method that combines multiple related tasks to learn based on a shared representation. It involves the parallel learning of multiple related tasks at the same time, and multiple tasks help each other learn through shallow shared representations to improve generalization. From Table 2, we can see that Tox21 pathway toxicity is a popular choice for multitask learning because of the information-sharing mechanism of its sub-datasets. In 2015, the research group of Thomas Unterthiner first applied multitask DL to toxicity prediction in a Tox21 data challenge and won both panel challenges (nuclear receptors and stress response) as well as the overall Grand Challenge, which ‘kicked off’ the use of DL to predict chemical toxicity. Given the high correlations between different subchallenges/tasks, the researchers concluded that the multitask learning approach significantly outperforms the single-task network in almost all tasks.^(p25) Recently, Liu *et al.* developed three multitask models for 12 Tox21 datasets based on a descriptor, FP, and molecular graph, respectively. Then, they proposed a comodel based on averaging for balance, and to make more accurate predictions, finally obtaining a comodel with an AUC of 0.903 and 0.897 for cross- and external validation, respectively.^(p96) Moreover, Das *et al.* have also demonstrated the advantage of employing a multitask model to predict toxicity across *in vitro*, *in vivo*, and clinical platforms and concluded that there is a minimal relative importance for *in vivo* data (animal data) for predicting clinical toxicity, in particular when unsupervised pretrained simplified

molecular input line entry system (SMILES) embeddings are used as input to multitask models.^(p97) Subsequent multitask research on some other toxicity endpoints has also supported this opinion. In 2019, Cheng *et al.* developed an effective prediction tool for drug-induced cardiotoxicity, termed deepHERG, based on 7889 compounds and a multitask DNN algorithm. Validation results showed that deepHERG outperformed models built using a single-task DNN, NB, SVM, RF, and graph convolutional neural network (GCNN). Specifically, the AUC value for the best deepHERG model was 0.967 on the validation set.^(p98) In 2021, Hou *et al.* proposed a novel multitask graph attention (MGA) framework to learn regression and classification tasks simultaneously and, using 33 toxicity datasets, confirmed the superiority of multitask prediction.^(p26) More recently, Martínez *et al.* built a novel model for the Ames test using a DL MTL approach and experimental information from five strains, and the results surpassed those obtained by single-task models.^(p32) Above all, these multitask studies have revealed the following advantages of the multitask model over the single-task model: (1) compared with single-task models, multitask models are more efficient, especially in the context of big data. (2) the multitask approach allows the capture of possible correlations between toxicity endpoints and the sharing of hidden unit representations among prediction tasks, which is particularly important for some tasks where very few measurements are available. Moreover, a further study based on 200 biological targets also found that adding additional tasks and data can improve the predictive power of a multitask DNN model.^(p99)

Although the advantage of DL models for predicting toxicity endpoints with big data is surprising, the improvement in model performance is very limited for those endpoints with small datasets. In the last few years, DL models have also been used to construct prediction models for some less-studied toxicity endpoints, but the results were usually less satisfactory or still inferior to TML due to their relatively small datasets.^{(p50),(p58),(p65)} In 2018, Tohkin *et al.* developed predictive classification models for hepatocellular hypertrophy using DL, RF, and SVM based on the Original Risk Assessment reports Database (ORAD), the Hazard Evaluation Support System Integrated Platform (HESS), and a combined dataset. The best model was found to be the SVM model using the HESS dataset, which was trained with 251 chemicals and predicted 214 test chemicals inside the application domain.^(p50) Similarly, in 2021, Li *et al.* collated a drug-induced ototoxicity dataset with 2807 molecules and constructed a series of prediction models using five TML and two DL algorithms. The results of internal and external validation showed that models based on SVM and RF performed better than two DL models, and a consensus model was finally obtained based on six TML models.^(p65) They drew a similar conclusion in a hematotoxicity prediction study conducted in the same year.^(p58) In these cases, DL algorithms did not show significant advantages compared with some outstanding TML methods, such as RF and SVM. The reason for this is that the main advantage of DL is embodied in big data, which can be used to automatically obtain high-level features using a large amount of

training data. Therefore we hypothesize that most DL algorithms will outperform traditional ML methods when more training data are available. In summary, although the application of DL in toxicity prediction research has been greatly advanced and achieved remarkable results, the main difficulty when using DL models for toxicity prediction is still insufficient training data.

The PDL

Given the difficulty posed by data sparsity in the field of data-driven toxicity prediction, and the fact that it is not always possible or easy to add more annotated data to a prediction task or construct multitask models, researchers in recent years have proposed some viable approaches to the ‘small data’ dilemma. To reduce the dependency of DL models on extensive toxic data, the semisupervised learning (SSL) method was applied to Tox21 by Siu *et al.* in 2021. SSL is a hybrid approach for learning tasks where only a portion of the data are labeled, and the amount of labeled data is much smaller than that of unlabeled data. It is a paradigm that can improve learning performance using a few labeled data and additional unlabeled examples as auxiliaries compared with supervised learning, and it provides a way to explore latent patterns through additional unlabeled examples, alleviating the need for a large number of labels.^(p100) SSL has found wide applications in various fields including bioinformatics, image classification, speech recognition, and many others.^(p101) In this study, the authors assumed that the label function is smooth in high-density areas, so data points located in the same area should share the same label. They proposed a GCN to predict chemical toxicity and trained the network via the mean teacher (MT) SSL algorithm. The results showed that the average AUC of the SSL model was 0.757, which is a 6% improvement over the model trained by supervised learning and conventional ML methods.^(p27) Encouraged by this success, SSL is expected to be beneficial to other chemical property prediction tasks through utilizing existing large unlabeled chemical databases.

Transfer learning (TL) is an ML approach that learns to perform a task in a data-rich environment and then ‘transfers’ the learned knowledge to a task with fewer available data. This approach is of great value for resolving situations where there is abundant labeled data for an associated problem but insufficient data for the problem under study.^(p102) Presently, TL is used to address low-data tasks in many fields such as computer vision,^(p103) natural language processing,^(p104) and drug discovery.^{(p105), (p106)} Current toxicity prediction studies involve many complex mechanisms and pathways, and TL thus has great potential to improve model performance in a DL environment. For example, Tung *et al.* developed an ensemble tree-based multitask learning method incorporating three relevant tasks in the well-defined AOP of skin sensitization to transfer shared knowledge to the major task of understanding human sensitizers. Their model showed superior performance compared with conventional single-task methods and the other four existing skin sensitization prediction methods.^(p107) Similarly, in 2020, Andrade *et al.* updated PredSkin to integrate multiple models developed with *in vitro*, *in chemico*, and mouse and human *in vivo* data, integrated into a consensus NB model that predicts human skin sen-

sitivity.^(p108) An integrative Bayesian model was trained on the ‘Human dataset’ using the predictions of models developed for five skin sensitization assays, described as descriptors, along with the NB algorithm and achieved a prediction accuracy of 0.89. In fact, TL is fundamentally different from SSL and multitask learning methods. Specifically, the source and target datasets of TL can have a different distribution and merely be related to each other, whereas in SSL, the source and target data are from the same dataset, and the target set does not have the labels.

Multitask learning aims to solve multiple tasks simultaneously and to learn a shared representation/model that performs well for all studied tasks. In contrast, in TL, the target domain is the focus, knowledge has already been obtained for target data from source data, and the domains do not need to be related or to function simultaneously.^(p109) Several examples have also shown that domain relatedness is more important than data size for TL.^(p110) In recent years, TL-related research has developed rapidly; although the application of TL is still in its infancy, and there is still much to be studied, it may soon be more effective and widely used in the context of small data. Furthermore, artificial data generation and RL may also be promising tools to mine useful information from small data.

In addition to SSL and TL, property augmentation is another effective approach to small data modeling. To take advantage of the expressive power of deep graph learning models, in 2021, Huang *et al.* proposed a property augmentation strategy to include massive training data along with other information to obtain a qualified DILI dataset for a DL algorithm. Based on the fact that there are overlapping molecules between initial DILI and hERG, Ames, mitochondrial membrane potential, and phospholipidosis datasets, the authors sought to utilize this additional toxicity information to enhance the prediction performance of DILI. Specifically, they combined these five datasets based on a SMILES representation of the drugs and then employed a multilabel training approach to establish a property augmentation learning process. Extensive experiments demonstrated that the proposed method significantly outperformed all existing baselines on the DILI datasets. More recently, Wang *et al.* also proposed a framework that combines feature selection, TL, and data augmentation for toxicity prediction and classification based on a small sample size. They obtained augmented data by adding white noise and then utilized these data together with the source domain data to train a CNN.^(p111) Property augmentation learning can provide more input data to better learn molecular vector representations and thus is a promising approach to improve the prediction performance for toxicity endpoints with limited data available.^(p86)

To sum up, recent toxicity prediction studies based on DL highlight the fact that DL can, and in fact already has, greatly impact the field of toxicity prediction with its powerful learning ability. It can not only generate high-quality single- and multitask prediction models but also discover new or potential structural alarms, which is a great help for medicinal chemists aiming to mine toxicity information from toxicity datasets. To solve the small data-related issues in the DL context, some novel methods including SSL, TL, and property augmentation learning have been proposed and achieved satisfactory results. Although there are still some difficulties in the application of DL, such as

TABLE 3
Detailed information on popular toxicity evaluation tools

	Endpoint	Methods	Accuracy	Toxic substructure	Explanation	URL
ADMETlab3.0	36 endpoints and 8 rules	DMPNN	0.67 ~ 0.97	Yes	+++	https://admetlab3.scbdd.com/
admetSAR3.0	55	CLMGraph	~0.87	No	+++	https://lmmd.ecust.edu.cn/admetSar3/
Deep-PK	21	DMPNN	0.16–0.89	Yes	+++	https://biosig.lab.uq.edu.au/deeppk/
HelixADMET	16	GNN	0.71 ~ 0.95	No	+	https://paddlehelix.baidu.com/app/drug/admet/train
pkCSM	10	GBS	0.68 ~ 0.90	No	++	https://biosig.unimelb.edu.au/pkcsM/prediction
vNN-ADMET	6	VNN	0.71 ~ 0.89	No	+	https://vnnadmet.bhsai.org/vnnadmet/login.xhtml
FP-ADMET	29	RF	0.64 ~ 0.89	No	+	https://gitlab.com/vishsoft/fpadmet
ICDrug	6	RF	–	No	+	https://www.icdrug.com/ICDrug/T
ProTox3.0	61	RF, DNN	0.75 ~ 0.93	No	+++	https://tox.charite.de
VenomPred2.0	8	RF, SVM, kNN, MLP	0.76 ~ 0.98	No	++	https://www.mmvsl.it/wp/venompred2/
AquaticTox	5	RF, AdaBoost, gradient boosting, SVM, FCNet, GACNN	0.71–0.8	No	+	https://chemyang.ccnu.edu.cn/ccb/server/AquaticTox
eMolTox	14	RF	0.83 ~ 0.94	Yes	++	https://xundrug.cn/moltox
TEST	9	Consensus	–	No	+	https://www.epa.gov/chemical-research/toxicity-estimation-software-tool-test
Toxtree	15 rules	DT	–	Yes	++	https://apps.ideaconsult.net/data/ui/toxtree
VirtualRat	12	RF, DT	–	No	+	https://virtualrat.cmdm.tw/
Pred-Skin 3.0	1	NB	0.89	No	+	https://predskin.labmol.com.br/
Pred-hERG 5.0	1	SVM	0.8	No	+	https://predherg.labmol.com.br/
CarcinoPred-EL	1	XGBoost	0.7	No	+	https://ccsibp.lnu.edu.cn/toxicity/CarcinoPred-EL/
CardPred	1	ANN	0.9	No	+	https://bioanalysis.cau.ac.kr:7050/
CardioDPi	3	DNN	0.89–0.94	No	+	https://cardiodpi.sapredictor.cn/

Abbreviations: CLMGraph, contrastive learning based multitask graph neural network; FCNet, fusion-correction network; GBS, graph-based signatures; VNN, variable nearest neighbor; GACNN, convolutional neural networks based on genetic algorithm.

the need for a sufficiently balanced and large dataset, the lack of transparency and interpretation, the long period over which the program must be run, and the need for a high-performance computer, etc., we believe that DL will bring new opportunities for toxicity prediction with the development of standard toxicity databases and updates in computer technology.

Prediction software and tools

Encouraged by the continuous accumulation of excellent prediction models for various toxicity endpoints and the urgent demand from toxicologists for practical research results, some prediction software and tools have been developed for single or multiple endpoints. In this part, we collated the freely available web tools and list their detailed information in Table 3.

There are 19 accessible tools for toxicity prediction detailed in Table 3, and their information is carefully summarized including modeling methods, prediction accuracy, and the interpretability of their prediction results. All the webserver tools can be classified into two categories according to the endpoints involved: comprehensive absorption, distribution, metabolism, excretion, and toxicity (ADMET) systems^{(p112),(p113),(p114),(p115),(p116),(p117),(p118),(p119)} and individual toxicity prediction tools.^{(p108),(p120),(p121),(p122),(p123),(p124),(p125),(p126),(p127),(p128),(p129),(p130)}

Similar to the evolutionary trends in toxicity endpoint prediction and modeling approaches, prediction tools have also become more practical as data have increased and methods have improved. Among the ADMET systems, the top ones are ADMETlab 3.0,^(p112) admetSAR 3.0,^(p113) Deep-PK,^(p114) and HelixADMET,^(p115) which have comparable prediction accuracy based on ML or DL (0.8 ~ 0.9) but differ in terms of the explanatory power of the prediction results. Typically, in ADMETlab 3.0 projects, the multitask DMPNN architecture coupled with molecular descriptors not only guarantees calculation speed for each endpoint simultaneously analyzed but also achieves superior performance in terms of accuracy and robustness. In addition to the basic prediction function, an application programming interface (API) has been introduced to meet the growing demand for programmatic access to large amounts of data in ADMETlab 3.0. To further improve the utility of the toxicity prediction tool, a new module was embedded in the web server of admetSAR 3.0, called ADMETopt, which can automatically replace other scaffolds in its scaffold library; this is very attractive to medicinal chemists who need to perform lead optimization. Deep-PK is a recently reported deep learn-based ADMET prediction platform that supports property prediction, molecular optimization, and interpretation to help users better understand the pharmacokinetics and toxicity of a given input molecule.^(p114) Recently, Wang *et al.* developed a highly robust and extensible ADMET system called HelixADMET, which incorporates the concept of self-supervised learning and a GNN. It can not only make accurate predictions even for molecules with unobserved scaffolds but also provides users with a model fine-tuning interface, allowing them to build new and customized endpoints. As for individual toxicity prediction tools, those with multiple endpoints are more practical, such as ProTox3.0,^(p122) eMolTox,^(p120) and AquaticTox.^(p124) They add some more complex endpoints (e.g. hepatotoxicity, nephrotoxicity, TOX-21 pathway toxicity) to the toxicity

repository and apply more reputable ML approaches (e.g. RF, SVM, and kNN) to obtain better predictive power (accuracy = 0.75 ~ 0.94). Moreover, some webserver tools focus on single but exceptionally important toxicity endpoints, such as CardPred^(p129) to predict carcinogenicity, Pred-hERG 5.0^(p127) to predict cardiotoxicity, and PredSkin 3.0^(p108) to predict skin sensitivity.

In summary, most of the best webserver tools for toxicity prediction are ADMET systems, which have improved and evolved with the addition of large databases and computer programs. Although existing software packages for making individual toxicity predictions already have considerable predictive power and practicability, they could be further improved by applying novel DL approaches to build more accurate models or building models predicting more toxicity endpoints. We believe that with the continuous development of new algorithms, the disclosure of standard toxicity datasets, and the progress of molecular representation methods, we will have access to more scientific and practical individual toxicity assessment tools in the near future.

Concluding remarks

Toxicity assessment is an important part of safety evaluation for new drugs, which is closely related to human health and the fate of drug candidates. In recent years, data-driven toxicity prediction has progressed from TML to DL, and then to the PDL. In this process, a series of remarkable prediction platforms have been generated, especially DL-based and multiple-endpoint toxicity prediction platforms with higher prediction accuracy and wider applicability. In this paper, we reviewed and discussed the features and difficulties of toxicity prediction, the evolution of modeling approaches, and the available predictive tools. Considering the research status of data-driven toxicity prediction, there are probably several directions for future toxicity prediction: (1) creating benchmark toxicity databases with sufficient and reliable experimental data; (2) constructing more elaborate or mechanistically explicit models, such as QSAR models based on known AOPs; (3) developing novel methods (e.g. new TL methods) to capture underlying information in small datasets more efficiently; (4) taking account of more interpretive or informative molecular attributes, such as descriptors that carry information about toxic targets; and (5) developing comprehensive toxicity prediction systems for various endpoints using advanced DLs. In conclusion, studies of toxicity prediction are updated annually, accompanied by improvements in data and computer programs. Although there are still some difficulties, we believe that data-driven toxicity prediction will soon move to a new stage and become an indispensable practical tool for drug safety assessment.

Funding

This work was supported by the National Key Research and Development Program of China (2021YFF1201400), the National Natural Science Foundation of China (22173118, 22220102001), the Hunan Provincial Science Fund for Distinguished Young Scholars (2021JJ10068), the Science and Technology Innovation Program of Hunan Province (2021RC4011), the Natural Science Foundation of Hunan Province (2023JJ60521,

2022JJ80104), and the 2020 Guangdong Provincial Science and Technology Innovation Strategy Special Fund (2020B1212030006, Guangdong-Hong Kong-Macau Joint Lab). Funding for open access was provided by the HKBU Strategic Development Fund (SDF19-0402-P02).

Declarations of interest

The authors declare no conflict of interest related to this study.

CRediT authorship contribution statement

Ningning Wang: Writing – original draft. **Xinliang Li:** Data curation. **Jing Xiao:** Data curation. **Shao Liu:** Conceptualization. **Dongsheng Cao:** Conceptualization.

Data availability

No data was used for the research described in the article.

References

- Paul SM et al. How to improve R&D productivity: the pharmaceutical industry's grand challenge. *Nat Rev Drug Discov.* 2010;9:203–214.
- Giri S, Bader A. A low-cost, high-quality new drug discovery process using patient-derived induced pluripotent stem cells. *Drug Discov Today.* 2015;20:37–49.
- Harrison RK. Phase II and Phase III failures: 2013–2015. *Nat Rev Drug Discov.* 2016;15:817–818.
- Smietana K, Siatkowski M, Moller M. Trends in clinical success rates. *Nat Rev Drug Discov.* 2016;15:379–380.
- Yang H, Sun L, Li W, Liu G, Tang Y. *In silico* prediction of chemical toxicity for drug design using machine learning methods and structural alerts. *Front Chem.* 2018;6:1–12.
- Lima AN, Philot EA, Trossini GH, Scott LP, Maltarollo VG, Honorio KM. Use of machine learning approaches for novel drug discovery. *Expert Opin Drug Discov.* 2016;11:225–239.
- Chen X, Xie D, Zhao Q, You ZH. MicroRNAs and complex diseases: from experimental results to computational models. *Brief Bioinform.* 2019;20:515–539.
- Raies AB, Bajic VB. *In silico* toxicology: computational methods for the prediction of chemical toxicity. *Wiley Interdiscip Rev Comput Mol Sci.* 2016;6:147–172.
- Dobchev DA, Pillai GG, Karelson M. *In silico* machine learning methods in drug development. *Curr Top Med Chem.* 2014;14:1913–1922.
- Segall MD, Barber C. Addressing toxicity risk when designing and selecting compounds in early drug discovery. *Drug Discov Today.* 2014;19:688–693.
- Hansch C, Maloney P, Fujita T, Muir RM. Correlation of biological activity of phenoxyacetic acids with hammett substituent constants and partition coefficients. *Nature.* 1962;194:178–180.
- Dearden JC. *In silico* prediction of drug toxicity. *J Comput Aided Mol Des.* 2003;17:119–127.
- Toropov AA, Toropova AP, Raska IJ, Leszczynska D, Leszczynski J. Comprehension of drug toxicity: software and databases. *Comput Biol Med.* 2014;45:20–25.
- Idakwo G et al. A review on machine learning methods for *in silico* toxicity prediction. *J Environ Sci Health C Environ Carcinog Ecotoxicol Rev.* 2018;36:169–191.
- Zhang L et al. Applications of machine learning methods in drug toxicity prediction. *Curr Top Med Chem.* 2018;18:987–997.
- Perkins R, Fang H, Tong W, Welsh WJ. Quantitative structure-activity relationship methods: perspectives on drug discovery and toxicology. *Environ Toxicol Chem.* 2003;22:1666–1679.
- Cherkasov A et al. QSAR modeling: where have you been? Where are you going to? *J Med Chem.* 2014;57:4977–5010.
- Tropsha A. Best practices for QSAR model development, validation, and exploitation. *Mol Inform.* 2010;29:476–488.
- Rachmadi MF, Valdes-Hernandez M, Agan M, Di Perri C, Komura T. Segmentation of white matter hyperintensities using convolutional neural networks with global spatial information in routine clinical brain MRI with none or mild vascular pathology. *Comput Med Imaging Graph.* 2018;66:28–43.
- Ma J, Sheridan RP, Liaw A, Dahl GE, Svetnik V. Deep neural nets as a method for quantitative structure-activity relationships. *J Chem Inf Model.* 2015;55:263–274.
- Ching T et al. Opportunities and obstacles for deep learning in biology and medicine. *J R Soc Interface.* 2018;15:1–47.
- Tang W, Chen J, Wang Z, Xie H, Hong H. Deep learning for predicting toxicity of chemicals: a mini review. *J Environ Sci Health C Environ Carcinog Ecotoxicol Rev.* 2018;36:252–271.
- Rim KT. *In silico* prediction of toxicity and its applications for chemicals at work. *Toxicol Environ Health Sci.* 2020;12:191–202.
- Kurosaki K, Wu R, Uesawa Y. A toxicity prediction tool for potential agonist/antagonist activities in molecular initiating events based on chemical structures. *Int J Mol Sci.* 2020;21:1–20.
- Unterthiner T, Mayr A, Klambauer G, Hochreiter S. Toxicity prediction using deep learning. *Comput Sci.* 2015;8:1–10.
- Wu Z, Jiang D, Wang J, Hsieh CY, Cao D, Hou T. Mining toxicity information from large amounts of toxicity data. *J Med Chem.* 2021;64:6924–6936.
- Chen J, Si YW, Un CW, Siu S. Chemical toxicity prediction based on semi-supervised learning and graph convolutional neural network. *J Cheminform.* 2021;13:1–16.
- Kalyaanamoorthy S, Barakat KH. Development of safe drugs: the hERG challenge. *Med Res Rev.* 2018;38:525–555.
- Cavalluzzi MM et al. Human ether-à-go-go-related potassium channel: exploring SAR to improve drug design. *Drug Discov Today.* 2020;25:344–366.
- Garrido A, Lepailleur A, Mignani SM, Dallemagne P, Rochais C. hERG toxicity assessment: useful guidelines for drug design. *Eur J Med Chem.* 2020;195, 112290.
- Wang S, Sun H, Liu H, Li D, Li Y, Hou T. ADMET evaluation in drug discovery. 16. Predicting hERG blockers by combining multiple pharmacophores and machine learning approaches. *Mol Pharm.* 2016;13:2855–2866.
- Martinez MJ et al. Multitask deep neural networks for Ames mutagenicity prediction. *J Chem Inf Model.* 2022;62:6342–6351.
- Honma M et al. Improvement of quantitative structure-activity relationship (QSAR) tools for predicting Ames mutagenicity: outcomes of the Ames/QSAR International Challenge Project. *Mutagenesis.* 2019;34:3–16.
- Benigni R, Bassan A, Pavan M. *In silico* models for genotoxicity and drug regulation. *Expert Opin Drug Metab Toxicol.* 2020;16:651–662.
- Herrmann K, Holzwarth A, Rime S, Fischer BC, Kneuer C. (Q)SAR tools for the prediction of mutagenic properties: are they ready for application in pesticide regulation? *Pest Manag Sci.* 2020;76:3316–3325.
- Ames BN, McCann J, Yamasaki E. Methods for detecting carcinogens and mutagens with the *Salmonella*/mammalian-microsome mutagenicity test. *Mutat Res.* 1975;31:347–364.
- Mortelmans K, Zeiger E. The Ames *Salmonella*/microsome mutagenicity assay. *Mutat Res.* 2000;455:29–60.
- Weibel HE, Kimber TB, Radetzki S, Neuenschwander M, Nazare M, Volkamer A. Revealing cytotoxic substructures in molecules using deep learning. *J Comput Aided Mol Des.* 2020;34:731–746.
- Riss TL, Moravec RA, Niles AL. Cytotoxicity testing: measuring viable cells, dead cells, and detecting mechanism of cell death. *Methods Mol Biol.* 2011;740:103–114.
- Mervin LH et al. Understanding cytotoxicity and cytostaticity in a high-throughput screening collection. *ACS Chem Biol.* 2016;11:3007–3023.
- Svensson F, Norinder U, Bender A. Modelling compound cytotoxicity using conformal prediction and PubChem HTS data. *Toxicol Res (Camb).* 2017;6:73–80.
- Walum E. Acute oral toxicity. *Environ Health Perspect.* 1998;106(Suppl 2):497–503.
- Lu J et al. Estimation of acute oral toxicity in rat using local lazy learning. *J Cheminform.* 2014;6:26–36.
- Lei T, Li Y, Song Y, Li D, Sun H, Hou T. ADMET evaluation in drug discovery: 15. Accurate prediction of rat oral acute toxicity using relevance vector machine and consensus modeling. *J Cheminform.* 2016;8:6–24.
- Hamadache M, Benkortbi O, Hanini S, Amrane A, Khaouane L, Si MC. A quantitative structure activity relationship for acute oral toxicity of pesticides on rats: validation, domain of application and prediction. *J Hazard Mater.* 2016;303:28–40.

46. Lavado GJ et al. Zebrafish AC₅₀ modelling: (Q)SAR models to predict developmental toxicity in zebrafish embryo. *Ecotoxicol Environ Saf.* 2020;202, 110936.
47. Marzo M et al. Integrating *in silico* models to enhance predictivity for developmental toxicity. *Toxicology.* 2016;370:127–137.
48. Patlewicz G, Ball N, Booth ED, Hulzebos E, Zvinavashe E, Hennes C. Use of category approaches, read-across and (Q)SAR: general considerations. *Regul Toxicol Pharmacol.* 2013;67:1–12.
49. Asilar E, Hemmerich J, Ecker GF. Image based liver toxicity prediction. *J Chem Inf Model.* 2020;60:1111–1121.
50. Ambe K et al. *In silico* prediction of chemical-induced hepatocellular hypertrophy using molecular descriptors. *Toxicol Sci.* 2018;162:667–675.
51. Liu L et al. Three-level hepatotoxicity prediction system based on adverse hepatic effects. *Mol Pharm.* 2019;16:393–408.
52. Fan D et al. *In silico* prediction of chemical genotoxicity using machine learning methods and structural alerts. *Toxicol Res (Camb).* 2018;7:211–220.
53. Benigni R, Bossa C, Worth A. Structural analysis and predictive value of the rodent *in vivo* micronucleus assay results. *Mutagenesis.* 2010;25:335–341.
54. Wichard JD. *In silico* prediction of genotoxicity. *Food Chem Toxicol.* 2017;106:595–599.
55. Jiang C, Zhao P, Li W, Tang Y, Liu G. *In silico* prediction of chemical neurotoxicity using machine learning. *Toxicol Res (Camb).* 2020;9:164–172.
56. Lei T et al. ADMET evaluation in drug discovery. 18. Reliable prediction of chemical-induced urinary tract toxicity by boosting machine learning approaches. *Mol Pharm.* 2017;14:3935–3953.
57. Cui X, Liu J, Zhang J, Wu Q, Li X. *In silico* prediction of drug-induced rhabdomyolysis with machine-learning models and structural alerts. *J Appl Toxicol.* 2019;39:1224–1232.
58. Hua Y, Shi Y, Cui X, Li X. *In silico* prediction of chemical-induced hematotoxicity with machine learning and deep learning methods. *Mol Divers.* 2021;25:1585–1596.
59. Long TZ et al. Structural analysis and prediction of hematotoxicity using deep learning approaches. *J Chem Inf Model.* 2023;63:111–125.
60. Mazzatorta P, Estevez MD, Coulet M, Schilter B. Modeling oral rat chronic toxicity. *J Chem Inf Model.* 2008;48:1949–1954.
61. Zhang C et al. *In silico* prediction of chemical toxicity on avian species using chemical category approaches. *Chemosphere.* 2015;122:280–287.
62. Zheng S, Xiong J, Wang Y, Liang G, Xu Y, Lin F. Quantitative prediction of hemolytic toxicity for small molecules and their potential hemolytic fragments by machine learning and recursive fragmentation methods. *J Chem Inf Model.* 2020;60:3231–3245.
63. Zhao P et al. *In silico* prediction of mitochondrial toxicity of chemicals using machine learning methods. *J Appl Toxicol.* 2021;41:1518–1526.
64. Xu X et al. *In silico* prediction of chemical acute contact toxicity on honey bees via machine learning methods. *Toxicol in Vitro.* 2021;72:105089–105095.
65. Huang X, Tang F, Hua Y, Li X. *In silico* prediction of drug-induced ototoxicity using machine learning and deep learning methods. *Chem Biol Drug Des.* 2021;98:248–257.
66. Cheng F et al. *In silico* assessment of chemical biodegradability. *J Chem Inf Model.* 2012;52:655–669.
67. Sun Y, Shi S, Li Y, Wang Q. Development of quantitative structure-activity relationship models to predict potential nephrotoxic ingredients in traditional Chinese medicines. *Food Chem Toxicol.* 2019;128:163–170.
68. Mulliner D, Schmidt F, Stolte M, Spirkel HP, Czich A, Amberg A. Computational models for human and animal hepatotoxicity with a global application scope. *Chem Res Toxicol.* 2016;29:757–767.
69. Huang R et al. Modelling the Tox21 10 K chemical profiles for *in vivo* toxicity prediction and mechanism characterization. *Nat Commun.* 2016;7:10425–10434.
70. Honma M. An assessment of mutagenicity of chemical substances by (quantitative) structure-activity relationship. *Genes Environ.* 2020;42:23–35.
71. Deng L, Hinton G, Kingsbury B. New types of deep neural network learning for speech recognition and related applications: an overview 2013 *IEEE international conference on acoustics, speech, and signal processing*:8599–8603.
72. LeCun Y, Bengio Y, Hinton G. Deep learning. *Nature.* 2015;521:436–444.
73. Bengio Y, Courville A, Vincent P. Representation learning: a review and new perspectives. *IEEE Trans Pattern Anal Mach Intell.* 2013;35:1798–1828.
74. Devillers J. Methods for building QSARs. *Methods Mol Biol.* 2013;930:3–27.
75. Omer A, Singh P, Yadav NK, Singh RK. An overview of data mining algorithms in drug induced toxicity prediction. *Mini Rev Med Chem.* 2014;14:345–354.
76. Cao Q et al. *In silico* estimation of chemical aquatic toxicity on crustaceans using chemical category methods. *Environ Sci Process Impacts.* 2018;20:1234–1243.
77. Cheng F, Li W, Liu G, Tang Y. *In silico* ADMET prediction: recent advances, current challenges and future trends. *Curr Top Med Chem.* 2013;13:1273–1289.
78. Nguyen-Vo TH et al. Predicting drug-induced liver injury using convolutional neural network and molecular fingerprint-embedded features. *ACS Omega.* 2020;5:25432–25439.
79. Hu Y, Ren Q, Liu X, Gao L, Xiao L, Yu W. *In silico* prediction of human organ toxicity via artificial intelligence methods. *Chem Res Toxicol.* 2023;36:1044–1054.
80. Ryu JY, Lee MY, Lee JH, Lee BH, Oh KS. DeepHIT: a deep learning framework for prediction of hERG-induced cardiotoxicity. *Bioinformatics.* 2020;36:3049–3055.
81. Siramshetty VB, Nguyen DT, Martinez NJ, Southall NT, Simeonov A, Zakharov AV. Critical assessment of artificial intelligence methods for prediction of hERG channel inhibition in the “Big Data” era. *J Chem Inf Model.* 2020;60:6007–6019.
82. Shan M, Jiang C, Chen J, Qin LP, Qin JJ, Cheng G. Predicting hERG channel blockers with directed message passing neural networks. *RSC Adv.* 2022;12:3423–3430.
83. Karim A, Lee M, Balle T, Sattar A. CardioTox Net: a robust predictor for hERG channel blockade based on deep learning meta-feature ensembles. *J Cheminform.* 2021;13:60.
84. Zhang X, Mao J, Wei M, Qi Y, Zhang J. HergSPred: accurate classification of hERG blockers/nonblockers with machine-learning models. *J Chem Inf Model.* 2022;62:1830–1839.
85. Hwang D, Jeon M, Kang J. A drug-induced liver injury prediction model using transcriptional response data with graph neural network 2020 *IEEE international conference on big data and smart computing (BigComp)*:323–329.
86. Ma H, An W, Wang Y, Sun H, Huang R, Huang J. Deep graph learning with property augmentation for predicting drug-induced liver injury. *Chem Res Toxicol.* 2021;34:495–506.
87. Gustavsson M et al. Transformers enable accurate prediction of acute and chronic chemical toxicity in aquatic organisms. *Sci Adv.* 2024;10:6669–6678.
88. Hung C, Gini G. QSAR modeling without descriptors using graph convolutional neural networks: the case of mutagenicity prediction. *Mol Divers.* 2021;25:1283–1299.
89. Li S et al. MutagenPred-GCNs: a graph convolutional neural network-based classification model for mutagenicity prediction with data-driven molecular fingerprints. *Interdiscip Sci.* 2021;13:25–33.
90. Fradkin P, Young A, Atanackovic L, Frey B, Lee LJ, Wang B. A graph neural network approach for molecule carcinogenicity prediction. *Bioinformatics.* 2022;38:i84–i91.
91. Li T, Tong W, Roberts R, Liu Z, Thakkar S. DeepCarc: deep learning-powered carcinogenicity prediction using model-level representation. *Front Artif Intell.* 2021;4:757780–757792.
92. Karim A et al. Quantitative toxicity prediction via meta ensembling of multitask deep learning models. *ACS Omega.* 2021;6:12306–12317.
93. Zhou Y et al. ToxMPNN: a deep learning model for small molecule toxicity prediction. *J Appl Toxicol.* 2024;44:953–964.
94. Huang Z, Lou S, Wang H, Li W, Liu G, Tang Y. AttentiveSkin: to predict skin corrosion/irritation potentials of chemicals via explainable machine learning methods. *Chem Res Toxicol.* 2024;37:361–373.
95. Jain S, Siramshetty VB, Alves VM, Muratov EN, Zakharov AV. Large-scale modeling of multispecies acute toxicity end points using consensus of multitask deep learning methods. *J Chem Inf Model.* 2021;61:653–663.
96. Yuan LY et al. Co-model for chemical toxicity prediction based on multi-task deep learning. *Mol Inform.* 2023;42:2200257–2200274.
97. Sharma B et al. Accurate clinical toxicity prediction using multi-task deep neural nets and contrastive molecular explanations. *Sci Rep.* 2023;13:4908–4923.
98. Cai C et al. Deep learning-based prediction of drug-induced cardiotoxicity. *J Chem Inf Model.* 2019;59:1073–1084.
99. Ramsundar B, Kearnes S, Riley P, Webster D, Konerding D, Pande V. Massively multitask networks for drug discovery. *Comput Sci.* 2015;2:1–27.
100. Hyun M, Jeong J, Kwak N. Class-imbalanced semi-supervised learning. *Mach Learn.* 2020;2:1–16.
101. Dou B et al. Machine learning methods for small data challenges in molecular science. *Chem Rev.* 2023;13:8736–8780.
102. Iman M, Rasheed K, Arabnia HR. A review of deep transfer learning and recent advancements. *Technologies.* 2022;40:1–14.
103. Kan M, Wu J, Shan S, Chen X. Domain adaptation for face recognition: targetize source domain bridged by common subspace. *Int J Comput Vision.* 2014;109:94–109.

104. Dai W, Xue GR, Yang Q, Yu Y. Co-clustering based classification for out-of-domain documents. *Proceedings of the 13th ACM SIGKDD international conference on knowledge discovery and data mining*:210–219.
105. Cai C et al. Transfer learning for drug discovery. *J Med Chem*. 2020;63:8683–8694.
106. Duan Y et al. Improved GNNs for log D7.4 prediction by transferring knowledge from low-fidelity data. *J Chem Inf Model*. 2023;63:2345–2359.
107. Tung CW, Lin YH, Wang SS. Transfer learning for predicting human skin sensitizers. *Arch Toxicol*. 2019;93:931–940.
108. Borba J et al. Pred-skin: a web portal for accurate prediction of human skin sensitizers. *Chem Res Toxicol*. 2021;34:258–267.
109. Zhuang F et al. *A comprehensive survey on transfer learning*. 2019;23:1–34.
110. Abbasi K, Poso A, Ghasemi J, Amanlou M, Masoudi-Nejad A. Deep transferable compound representation across domains and tasks for low data drug discovery. *J Chem Inf Model*. 2019;59:4528–4539.
111. Zhao H, Qiu S, Bai M, Wang L, Wang Z. Toxicity prediction and classification of Gunqile-7 with small sample based on transfer learning method. *Comput Biol Med*. 2024;173:108348–108358.
112. Fu L et al. ADMETlab 3.0: an updated comprehensive online ADMET prediction platform enhanced with broader coverage, improved performance, API functionality and decision support. *Nucleic Acids Res*. 2024;52:W422–W431.
113. Gu Y et al. AdmetSAR3.0: a comprehensive platform for exploration, prediction and optimization of chemical ADMET properties. *Nucleic Acids Res*. 2024;52:W432–W438.
114. Myung Y, de Sa A, Ascher DB. Deep-PK: deep learning for small molecule pharmacokinetic and toxicity prediction. *Nucleic Acids Res*. 2024;52:W469–W475.
115. Zhang S et al. HelixADMET: a robust and endpoint extensible ADMET system incorporating self-supervised knowledge transfer. *Bioinformatics*. 2022;38:3444–3453.
116. Pires DE, Blundell TL, Ascher DB. PkCSM: predicting small-molecule pharmacokinetic and toxicity properties using graph-based signatures. *J Med Chem*. 2015;58:4066–4072.
117. Schyman P, Liu R, Desai V, Wallqvist A. VNN web server for ADMET predictions. *Front Pharmacol*. 2017;8:889–902.
118. Venkatraman V. FP-ADMET: a compendium of fingerprint-based ADMET prediction models. *J Cheminform*. 2021;13:75–86.
119. Yifei Qi JZHZ. ICDrug. <http://www.icdrug.com/ICDrug/T>. Accessed 7 December 2023.
120. Ji C, Svensson F, Zoufir A, Bender A. EMolTox: prediction of molecular toxicity with confidence. *Bioinformatics*. 2018;34:2508–2509.
121. EPA (2021). Toxicity estimation software tool [computer software].
122. Banerjee P, Kemmler E, Dunkel M, Preissner R. ProTox 3.0: a webserver for the prediction of toxicity of chemicals. *Nucleic Acids Res*. 2024;52:W513–W520.
123. Di Stefano M et al. VenomPred 2.0: a novel *in silico* platform for an extended and human interpretable toxicological profiling of small molecules. *J Chem Inf Model*. 2024;64:2275–2289.
124. Shi X, Wang ZZ, Wang YL, Wang F, Yang GF. AquaticTox: a web-based tool for aquatic toxicity evaluation based on ensemble learning to facilitate the screening of green chemicals. *Environ Health*. 2024;2:202–211.
125. Patlewicz G, Jeliaskova N, Safford RJ, Worth AP, Aleksiev B. An evaluation of the implementation of the Cramer classification scheme in the Toxtree software. *SAR QSAR Environ Res*. 2008;19:495–524.
126. University NT. VirtualRat. <https://virtualrat.cmdm.tw/>. Accessed 12 July 2023.
127. Braga RC et al. Pred-hERG: a novel web-accessible computational tool for predicting cardiac toxicity. *Mol Inform*. 2015;34:698–701.
128. Zhang L et al. CarcinoPred-EL: novel models for predicting the carcinogenicity of chemicals using molecular fingerprints and ensemble learning methods. *Sci Rep*. 2017;7:2118–2131.
129. Lee HM et al. Computational determination of hERG-related cardiotoxicity of drug candidates. *BMC Bioinformatics*. 2019;20:250–256.
130. Chen Z, Li N, Zhang P, Li Y, Li X. CardioDPi: an explainable deep-learning model for identifying cardiotoxic chemicals targeting hERG, Cav1.2, and Nav1.5 channels. *J Hazard Mater*. 2024;474:134724–134734.